Suggested Answers to Problem Set 1

1. $E(y_t) = 1$, and

$$\gamma_0 = Var(\varepsilon_t) + 0.6^2 Var(\varepsilon_{t-1}) + 0.2^2 Var(\varepsilon_{t-2}) = 1.4,$$

$$\begin{aligned}\gamma_1 &= E[(y_t - 1)(y_{t-1} - 1)] \\ &= E[(\varepsilon_t + 0.6\varepsilon_{t-1} + 0.2\varepsilon_{t-2})(\varepsilon_{t-1} + 0.6\varepsilon_{t-2} + 0.2\varepsilon_{t-3})] \\ &= 0.6E(\varepsilon_{t-1}^2) + 0.2 \times 0.6E(\varepsilon_{t-2}^2) \\ &= 0.6 + 0.12 = 0.72,\end{aligned}$$

$$\begin{aligned}\gamma_2 &= E[(y_t - 1)(y_{t-2} - 1)] \\ &= E[(\varepsilon_t + 0.6\varepsilon_{t-1} + 0.2\varepsilon_{t-2})(\varepsilon_{t-2} + 0.6\varepsilon_{t-3} + 0.2\varepsilon_{t-4})] \\ &= 0.2E(\varepsilon_{t-2}^2) = 0.2,\end{aligned}$$

$$\gamma_h = 0, \quad \text{for } h \geq 3.$$

It follows that $\rho_0 = 1$, $\rho_1 = 0.72/1.4 \approx 0.51$, $\rho_2 = 0.2/1.4 \approx 0.14$, and $\rho_h = 0$, for $h \geq 3$.

2. (a) No, y_t not weakly stationary. It is clear that $E(y_t) = \beta_0 + \beta_1 t$, which is time-varying. There is no need to further check the variance and autocovariance.

(b) Yes, x_t is weakly stationary. To see this,

$$\begin{aligned}x_t &= y_t - y_{t-1} \\ &= (\beta_0 + \beta_1 t + \varepsilon_t + \theta\varepsilon_{t-1}) - [\beta_0 + \beta_1(t-1) + \varepsilon_{t-1} + \theta\varepsilon_{t-2}] \\ &= \beta_1 + \varepsilon_t + (\theta - 1)\varepsilon_{t-1} - \theta\varepsilon_{t-2},\end{aligned}$$

which is an $MA(2)$ model, and we know all finite-order MA models are weakly stationary. To further verify the weak stationarity of this series, it is clear that $E(x_t) = \beta_1$, and for any t and h (assuming $h \geq 0$), we have

$$\begin{aligned}Cov(x_t, x_{t-h}) &= E[(x_t - \beta_1)(x_{t-h} - \beta_1)] \\ &= E\{[\varepsilon_t + (\theta - 1)\varepsilon_{t-1} - \theta\varepsilon_{t-2}][\varepsilon_{t-h} + (\theta - 1)\varepsilon_{t-h-1} - \theta\varepsilon_{t-h-2}]\},\end{aligned}$$

which equals zero if $h > 2$. If $h = 0$, we obtain $Var(x_t) = \left[1 + (\theta - 1)^2 + \theta^2\right]\sigma^2 =$

$2\left(\theta^2-\theta+1\right)\sigma^2$, which is a constant. If $h=1$, then

$$\begin{aligned} Cov\left(x_t, x_{t-1}\right) &= E\left\{\left[\varepsilon_t+\left(\theta-1\right)\varepsilon_{t-1}-\theta\varepsilon_{t-2}\right]\left[\varepsilon_{t-1}+\left(\theta-1\right)\varepsilon_{t-2}-\theta\varepsilon_{t-3}\right]\right\} \\ &= \left[\left(\theta-1\right)-\theta\left(\theta-1\right)\right]\sigma^2 = \left(-\theta^2+2\theta-1\right)\sigma^2. \end{aligned}$$

If $h=2$,

$$\begin{aligned} Cov\left(x_t, x_{t-2}\right) &= E\left\{\left[\varepsilon_t+\left(\theta-1\right)\varepsilon_{t-1}-\theta\varepsilon_{t-2}\right]\left[\varepsilon_{t-2}+\left(\theta-1\right)\varepsilon_{t-3}-\theta\varepsilon_{t-4}\right]\right\} \\ &= -\theta\sigma^2. \end{aligned}$$

In sum, given the parameter values, $Cov\left(x_t, x_{t-h}\right)$ is only a function of h and it does not depend on t.

3.

(a) By backward substitution,

$$\begin{aligned} \varepsilon_t &= -\theta\varepsilon_{t-1}+y_t \\ &= -\theta\left(-\theta\varepsilon_{t-2}+y_{t-1}\right)+y_t \\ &= \theta^2\left(-\theta\varepsilon_{t-3}+y_{t-2}\right)-\theta y_{t-1}+y_t \\ &= -\theta^3\varepsilon_{t-3}+\theta^2 y_{t-2}-\theta y_{t-1}+y_t \\ &\vdots \\ &= \left(-\theta\right)^{k+1}\varepsilon_{t-k-1}+\left(-\theta\right)^k y_{t-k}+\left(-\theta\right)^{k-1} y_{t-k+1}+\ldots+y_t, \end{aligned}$$

where the first term decays to zero as $k\to\infty$. Therefore,

$$\varepsilon_t = \sum_{j=0}^{\infty}(-\theta)^j y_{t-j}.$$

It is clear that the weight on y_{t-j} is $\left|(-\theta)^j\right|$, which decreases with j. Alternatively, we can use the lag operator and write the model as

$$y_t = \left(1+\theta L\right)\varepsilon_t.$$

Since $|\theta| < 1$, we have

$$\begin{aligned}\varepsilon_t &= (1+\theta L)^{-1} y_t = [1-(-\theta) L]^{-1} y_t \\ &= \left[1+(-\theta L)+(-\theta L)^2+(-\theta L)^3+\ldots\right] y_t \\ &= y_t - \theta y_{t-1} + \theta^2 y_{t-2} - \theta^3 y_{t-3} + \ldots \\ &= \sum_{j=0}^{\infty} (-\theta)^j y_{t-j}.\end{aligned}$$

Note that $(1+\theta L)^{-1} = 1+(-\theta L)+(-\theta L)^2+(-\theta L)^3+\ldots$ holds only if $|\theta| < 1$; otherwise, the infinite series is divergent.

(b) By forward substitution,

$$\begin{aligned}\varepsilon_t &= \theta^{-1}\varepsilon_{t+1} + \theta^{-1} y_{t+1} \\ &= \theta^{-1}\left(\theta^{-1}\varepsilon_{t+2} + \theta^{-1} y_{t+2}\right) + \theta^{-1} y_{t+1} \\ &= \theta^{-2}\left(\theta^{-1}\varepsilon_{t+3} + \theta^{-1} y_{t+3}\right) + \theta^{-2} y_{t+2} + \theta^{-1} y_{t+1} \\ &= \theta^{-3}\varepsilon_{t+3} + \theta^{-3} y_{t+3} + \theta^{-2} y_{t+2} + \theta^{-1} y_{t+1} \\ &\vdots \\ &= \theta^{-k}\varepsilon_{t+k} + \theta^{-k} y_{t+k} + \theta^{-k+1} y_{t+k-1} + \ldots + \theta^{-1} y_{t+1},\end{aligned}$$

where the first term, under $|\theta| > 1$, diminishes to zero as $k \to \infty$. Therefore

$$\varepsilon_t = \sum_{j=1}^{\infty} \theta^{-j} y_{t+j}.$$

4. Note that $z_t = \varepsilon_t - 0.4\varepsilon_{t-1} + u_t - 0.7u_{t-1}$. Clearly, z_t has zero mean and its autocovariances are given by

$$\begin{aligned}\gamma_0 &= E\left(z_t^2\right) = E\left(\varepsilon_t - 0.4\varepsilon_{t-1} + u_t - 0.7u_{t-1}\right)^2 \\ &= \sigma_\varepsilon^2 + 0.4^2\sigma_\varepsilon^2 + \sigma_u^2 + 0.7^2\sigma_u^2 = 1.16 \times 4 + 1.49 \times 1 = 6.13,\end{aligned}$$

$$\begin{aligned}\gamma_1 &= E\left(z_{t-1} z_t\right) = E\left[\left(\varepsilon_{t-1} - 0.4\varepsilon_{t-2} + u_{t-1} - 0.7u_{t-2}\right)\left(\varepsilon_t - 0.4\varepsilon_{t-1} + u_t - 0.7u_{t-1}\right)\right] \\ &= -0.4\sigma_\varepsilon^2 - 0.7\sigma_u^2 = -0.4 \times 4 - 0.7 \times 1 = -2.3,\end{aligned}$$

and $\gamma_h = 0$, for $h \geq 2$. Therefore z_t is stationary. Since $\gamma_h = 0$, for $h \geq 2$, z_t should be represented by an $MA(1)$ process. Suppose $z_t = e_t + \theta e_{t-1}$, where $e_t \sim WN\left(0, \sigma_e^2\right)$. We need

to find θ and σ_e^2 such that z_t has the same autocovariances as derived above. That is,

$$\gamma_0 = \left(1+\theta^2\right)\sigma_e^2 = 6.13$$
$$\gamma_1 = \theta\sigma_e^2 = -2.3.$$

Solving these two equations yield

$$\frac{\theta}{1+\theta^2} = \frac{-2.3}{6.13} = -0.375$$

$\theta = \frac{-1\pm\sqrt{1-4\times 0.375^2}}{2\times 0.375} = -2.215$ and -0.451. We choose the solution satisfying $|\theta| < 1$ to obtain an invertible MA process. Hence, $\theta = -0.451$ and $\sigma_e^2 = 5.10$.

5.

(a) $\left(1 - L + 0.24L^2\right) y_t = \varepsilon_t$.

(b) The characteristic equation

$$\lambda^2 - \lambda + 0.24 = (\lambda - 0.4)(\lambda - 0.6) = 0.$$

has roots $\lambda = 0.4$ and $\lambda = 0.6$, both lying inside the unit circle. Therefore, y_t is a stationary process. Since both roots are real, business cycles are not present.

(c)

$$(1 - 0.4L)(1 - 0.6L)\, y_t = \varepsilon_t$$

$$\begin{aligned} y_t &= (1-0.4L)^{-1}(1-0.6L)^{-1}\varepsilon_t \\ &= \left(1 + 0.4L + 0.4^2L^2 + \ldots\right)\left(1 + 0.6L + 0.6^2L^2 + \ldots\right)\varepsilon_t \\ &= \varepsilon_t + (0.4+0.6)\,\varepsilon_{t-1} + \left(0.4^2 + 0.4\times 0.6 + 0.6^2\right)\varepsilon_{t-2} + \ldots \end{aligned}$$

$$\psi_0 = 1,\ \psi_1 = 1,\ \psi_2 = 0.76,\ \ldots$$

It can be seen that

$$\psi_i = \sum_{j=0}^{i} 0.4^j 0.6^{i-j},\ \text{for } i = 0, 1, 2, \ldots.$$

Note that ψ_i satisfies the same second-order difference equation as y_t, namely, $\psi_i = \psi_{i-1} - 0.24\psi_{i-2}$, for $i = 2, 3, \ldots$.

(d) The mean of x_t is $\frac{2}{1-1+0.24} = 8.33$. Then

$$x_t - 8.33 = (x_{t-1} - 8.33) - 0.24\,(x_{t-2} - 8.33) + \varepsilon_t.$$

Let $y_t = x_t - 8.33$. We have shown in part (b) that $y_t = \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}$, with $\psi_i = \sum_{j=0}^{i} 0.4^j 0.6^{i-j}$. Therefore,

$$x_t = 8.33 + y_t = 8.33 + \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i},$$

where ψ_i's are the same as in part (c).

(e)

$$\lambda^2 - \lambda + 0.3 = 0$$

has roots

$$\lambda = \frac{1 \pm i\sqrt{4 \times 0.3 - 1^2}}{2} = 0.5 \pm 0.22i.$$

$R = \sqrt{0.5^2 + 0.22^2} = 0.55 < 1$. Hence, z_t is stationary. The average length of the stochastic cycles is

$$k = \frac{2\pi}{cos^{-1}(0.5/0.55)} \approx 14.6 \text{ periods.}$$

Question 6

(a) There are 2515 observations on 6 variables.

(b) It is clear from both time series and ACF plots that the price series is not stationary. The first four lagged ACFs are 0.997, 0.994, 0.991, and 0.988. They are all very close to 1 and highly significant. There undoubtedly exists strong serial dependence in the share price.

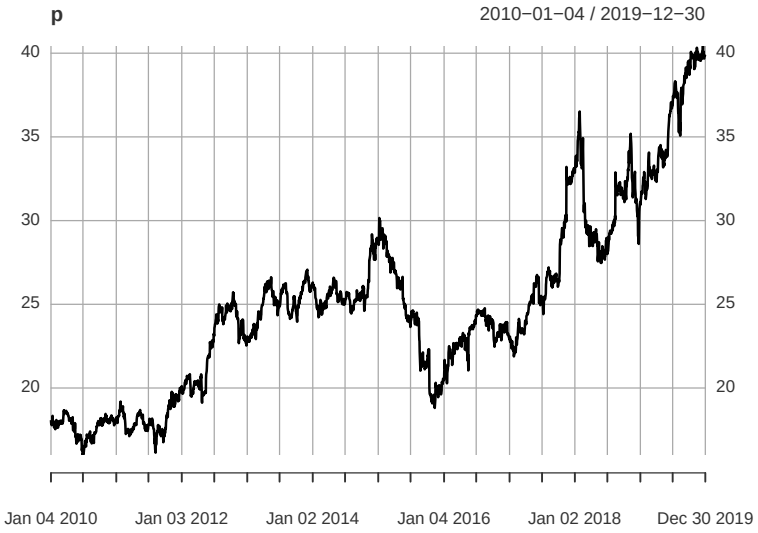
```
library(quantmod)
getSymbols("WMT", src="yahoo", from = "2010-01-01", to = "2019-12-31")
```

```
## [1] "WMT"
```

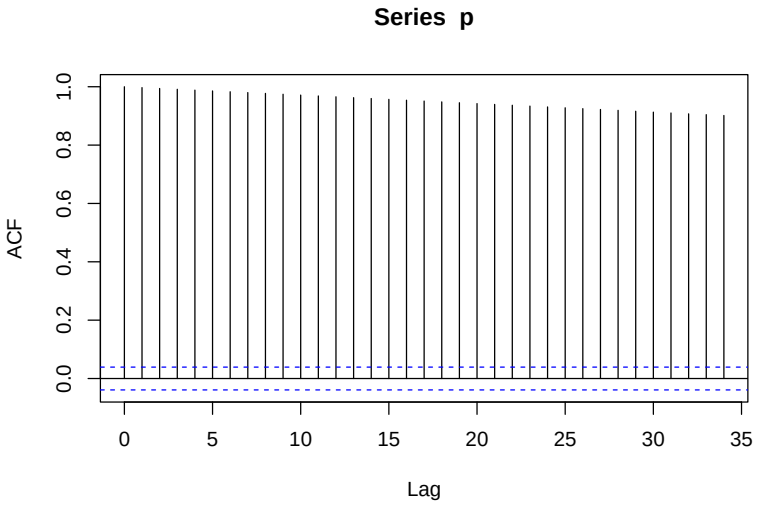
```
dim(WMT)
```

```
## [1] 2515    6
```

```
p <-WMT$WMT.Close
plot(p)
```




```
(acf(p))
```

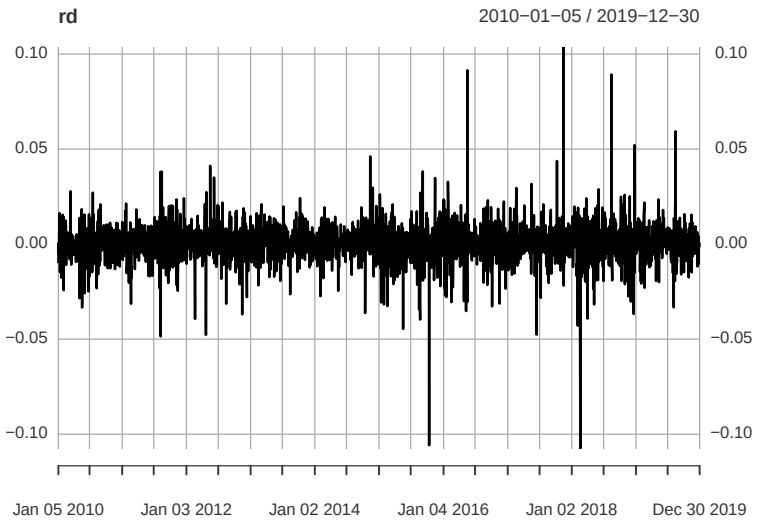



```
##
## Autocorrelations of series 'p', by lag
```

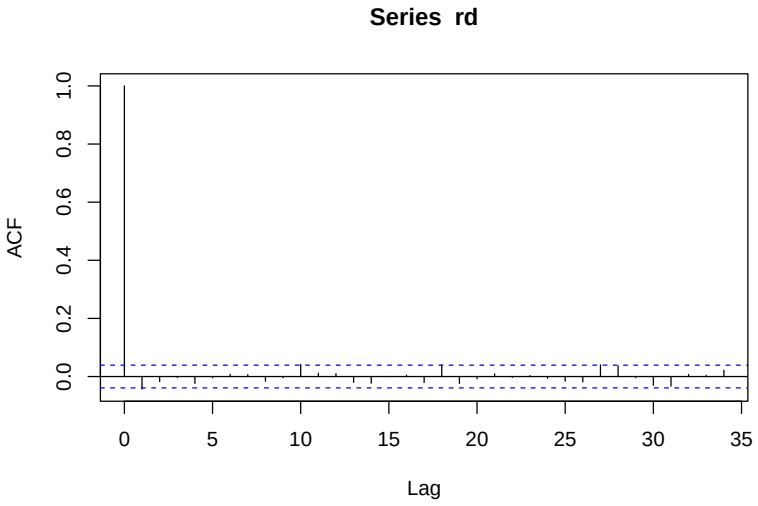
```
##
##     0     1     2     3     4     5     6     7     8     9    10    11    12
## 1.000 0.997 0.994 0.991 0.988 0.986 0.983 0.980 0.977 0.974 0.971 0.968 0.966
##    13    14    15    16    17    18    19    20    21    22    23    24    25
## 0.963 0.960 0.957 0.954 0.951 0.948 0.945 0.942 0.939 0.937 0.934 0.931 0.928
##    26    27    28    29    30    31    32    33    34
## 0.925 0.922 0.919 0.916 0.913 0.910 0.907 0.904 0.902
```

(c) Both time and ACF plots show that the daily return series look like stationary (although there are spikes in the variances). The first four lagged ACFs are -0.043, -0.017, -0.003, and -0.024. They are all very small and insignificant. (The lag-1 ACF is marginally significant but negative.)

```
rd <- diff(log(p))[-1]
plot(rd)
```




```
(acf(rd))
```




```
##
## Autocorrelations of series 'rd', by lag
##
##      0      1      2      3      4      5      6      7      8      9     10
##  1.000 -0.043 -0.017 -0.003 -0.024 -0.004  0.008  0.006 -0.016 -0.004  0.042
##     11     12     13     14     15     16     17     18     19     20     21
```

```
##  0.011  0.010 -0.020 -0.023  0.000  0.005 -0.021  0.040 -0.024 -0.007  0.009
##     22     23     24     25     26     27     28     29     30     31     32
## -0.003  0.003 -0.006 -0.016 -0.019  0.040  0.037 -0.004 -0.030 -0.033  0.007
##     33     34
##  0.004  0.022
```

(d) The first 5 observations and summary statistics are printed below.

```
head(rd, 5)
```

```
##               WMT.Close
## 2010-01-05 -0.010007513
## 2010-01-06 -0.002237606
## 2010-01-07  0.000559871
## 2010-01-08 -0.005050052
## 2010-01-11  0.016366353
```

```
cat(" Mean:", round(mean(rd), 6), "\n",
    "Median:", round(median(rd), 6), "\n",
    "Standard Deviation:", round(sd(rd), 6))
```

```
##  Mean: 0.000314
##  Median: 0.00057
##  Standard Deviation: 0.010895
```

```
colnames(rd) <-"rd"
```

(e) We have 522 and 120 observations on rw and rm, respectively. The null hypothesis of the Ljung-Box test is $H_0 : \rho_1 = \rho_2 = \ldots = \rho_{10} = 0$, and the alternative hypothesis is H_1 : At least one of ρ_h is nonzero, for $h = 1, 2, \ldots, 10$. The p-value of the test using daily, weekly, and monthly returns are 0.26, 0.12, and 0.73, respectively. We fail to reject the null hypothesis for all three tests, and therefore conclude that daily, monthly, and weekly returns are white noise processes (serially uncorrelated) even at the 10% level.

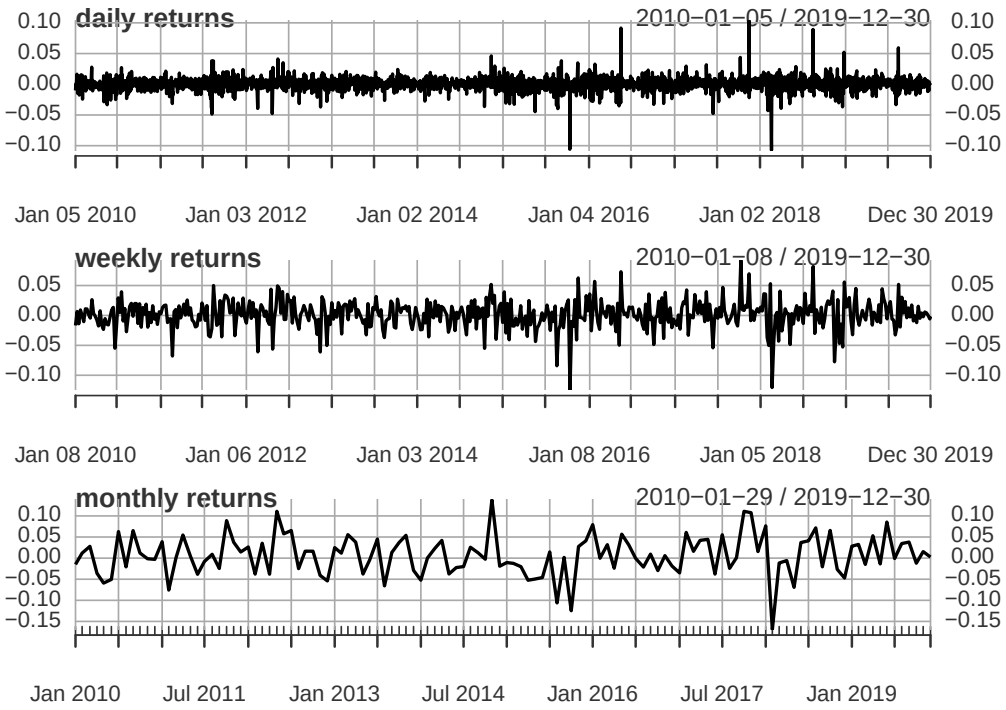
```
rw <- apply.weekly(rd, FUN = sum)
rm <- apply.monthly(rd, FUN = sum)
dim(rw)
```

```
## [1] 522   1
```

```
dim(rm)
```

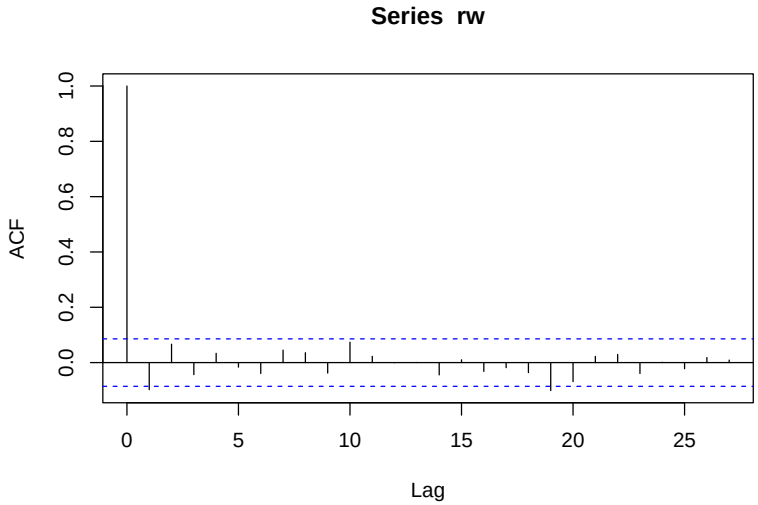
```
## [1] 120   1
```

```
par(mfrow=c(3,1))
plot(rd, main = "daily returns")
plot(rw, main = "weekly returns")
plot(rm, main = "monthly returns")
```

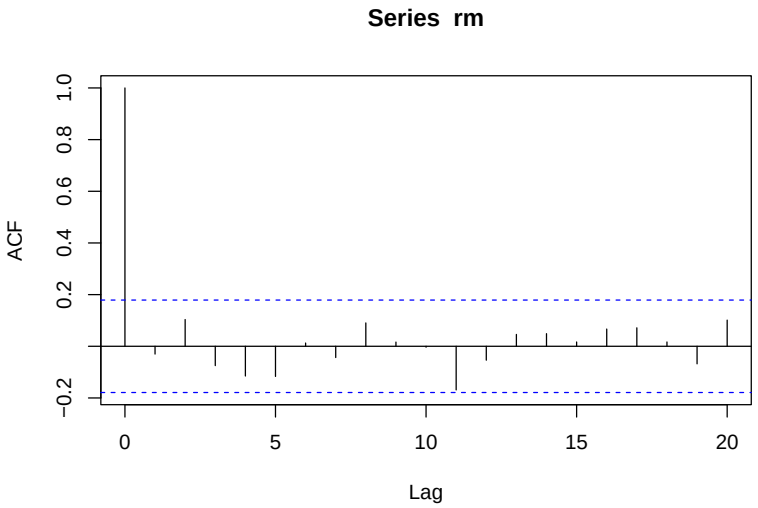



```
par(mfrow=c(1,1))  # restore default
```

```
acf(rw)
```




```
acf(rm)
```




```
Box.test(rd, lag = 10, type="Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  rd
## X-squared = 12.329, df = 10, p-value = 0.2636
```

```
Box.test(rw, lag = 10, type="Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  rw
## X-squared = 15.358, df = 10, p-value = 0.1195
```

```
Box.test(rm, lag = 10, type="Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  rm
## X-squared = 6.9229, df = 10, p-value = 0.7327
```

(f) We find no significant autocorrelations in returns at daily, weekly, or monthly frequencies, lending support to the hypothesis that returns of a financial asset are uncorrelated over time. As expected, daily returns have the smallest magnitude and exhibit greater volatility. You may also notice the phenomenon of volatility clustering in daily and weekly returns, where large (or small) absolute returns tend to occur in clusters. In general, daily returns are likely to exhibit a small degree of autocorrelation, which did not occur in this case (though we only examined the first 10 lags). I encourage you to explore stock returns of a few different companies and time periods.